

EKAMVEER SINGH

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SUMMARY

Second-year B.Tech student in Computer Science and Artificial Intelligence with a focus on efficient on-device ML inference and privacy-preserving AI systems. Building production-grade local AI pipelines from scratch on Apple Silicon, progressing from low-level tensor operations to a working voice-to-LLM assistant pipeline. Actively contributing to interdisciplinary AI research alongside PhD scholars and faculty.

EDUCATION

B.Tech — Computer Science and Artificial Intelligence (CGPA: 8.77)

Expected May 2028

Punjab Engineering College – Chandigarh, India

Minor — Foundations of Quantum Technology (CGPA: 10.00)

Expected May 2028

Punjab Engineering College – Chandigarh, India

SKILLS

Languages: Python, C++, Swift, SQL

ML / AI: MLX, MLX-VLM, PyTorch, TensorFlow, Scikit-learn, CatBoost, OpenAI Whisper, OpenCV

On-Device & Systems: MLX Swift, SwiftUI, Apple Silicon inference, Piper TTS, whisper.cpp, PyAudio

Data & Visualisation: NumPy, Pandas, Matplotlib, Seaborn, Plotly, MATLAB

Tools & Infra: Git, GitHub, Docker, VSCode, AWS, Ollama, Kaggle API, OpenAlex API

Databases: MySQL, PostgreSQL, MongoDB, SQLite

PROJECTS

Apple Silicon AI — Local Personal AI Assistant

2026-Present

github.com/Cheese-Singh — [MLX Series — Side Projects](#)

- Building a fully local, privacy-preserving AI assistant for Apple Silicon devices (MacBook M3, iPhone, Apple Watch) with zero cloud dependency.
- Implemented a progressive 4-stage pipeline: tensor fundamentals and manual embeddings (A1) → local LLM inference with Qwen3.5-9B via MLX (A2) → stateful multi-turn chat (A3) → live voice assistant (A4) recording audio, transcribing with OpenAI Whisper, and generating responses through MLX-VLM.
- Stack: mlx, mlx-vm, transformers, PyAudio, OpenAI Whisper, quantized Apple Silicon models. All code written independently, every stage committed separately.

Local LLM Inference & Agentic Tool-Use Series

2026

github.com/Cheese-Singh — [Ollama Series — Side Projects](#)

- Built a 5-script progressive series covering single-turn generation, multimodal vision-language input, stateful multi-turn chat with think-block stripping and repetition prevention, and an agentic function-calling loop.
- Agentic script (B4) implements a full tool-use loop: LLM reasons over a CSV-backed inventory, calls structured functions (check, add, update, save), handles errors, and executes multi-step plans autonomously.
- Final script (B5) builds a CLI for Modelfile lifecycle management, model validation, and automated prompt benchmarking with structured report generation.
- Stack: ollama, pandas, Python.

Research ML — Materials Science Prediction & Trend Analysis

2025

github.com/Cheese-Singh — [Research ML — Side Projects](#)

- Built a CatBoost regression pipeline to predict supercapacitor capacitance (F/g) from material, synthesis, and electrolyte features; evaluated with Leave-One-Out Cross-Validation and generated diagnostic plots (actual-vs-predicted, residual distribution, feature importance).
- Built a research trend analyser querying the OpenAlex API to track yearly publication volume for any topic and output bar chart visualisations.
- Stack: CatBoost, scikit-learn, pandas, NumPy, Matplotlib, OpenAlex API.

- A structured series of standalone scripts each isolating one classical ML concept: multi-classifier comparison (KNN, SVM, Decision Tree), SVM decision boundary visualisation across kernels and C values, automated EDA pipeline generating correlation heatmaps and histograms for any CSV.
- Includes a full sklearn preprocessing pipeline (Pipeline + ColumnTransformer) with imputation, scaling, and encoding — zero manual cleaning — demonstrated on the Titanic dataset.
- PCA vs. t-SNE interactive comparison on Fashion-MNIST using Plotly, highlighting the difference between linear and nonlinear dimensionality reduction.

RESEARCH & PUBLICATIONS

Co-author — "Current Trends in MXenes and their Hybrids for Next-Generation Supercapacitor Applications"

Submitted to Royal Society of Chemistry (2026). Interdisciplinary review written with a PhD scholar under faculty supervision.

Independently authored the AI methodology section, building ML regression models to analyse electrochemical performance data.

EXPERIENCE

Research Intern

Mar 2025 – Jul 2025

Punjab Engineering College – Chandigarh, India

- Conducted literature reviews for an interdisciplinary AI + materials science project spanning supercapacitor electrochemistry and ML-based performance prediction.
- Independently authored the AI methodology section of a co-authored review article submitted to RSC; implemented CatBoost regression pipeline with LOOCV evaluation.
- Collaborated with a PhD scholar and faculty from Computer Science and Physics departments.

ACCOMPLISHMENTS

- JEE Mains 2024: 98.8 percentile
- Published: "Solar-Powered Mobile Phones: The Future of Sustainable Energy", The Tribune (2023)
- Third Place, Intra-college Robosoccer Competition
- Participant, Smart India Hackathon (SIH) 2024

CLUBS & ACTIVITIES

- Technical Society Member — ACM Student Chapter & Robotics Club (Aug 2024 – Present)
- Counselling Cell Member (Aug 2024 – Present)
- NGO Volunteer — Aashray Foundation (Mar – Dec 2025)
- Football (Soccer) Team Player

LANGUAGES

Punjabi (Native) **English** (C2 — Proficient) **Hindi** (C2 — Proficient) **Norwegian** (B1 — Intermediate)